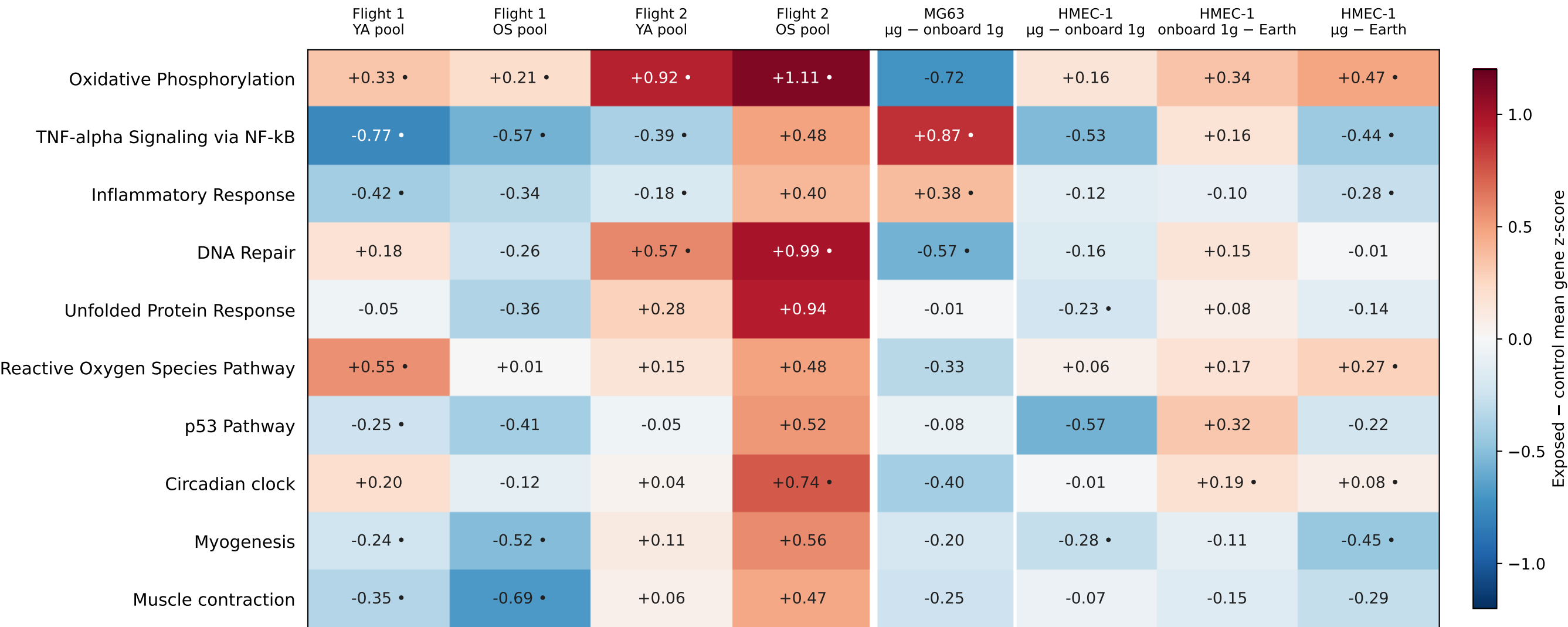


Narrow gene-set patterns survive a broader mixed response

All 10 predefined gene sets • human flight comparisons • exploratory, no significance tests



• Direction survives every sample deletion, rank/background score checks, and source-processing checks.
Dots are robustness flags, not statistical significance. Scores are standardized within each contrast.
Muscle chips: 3 + 3 per pool/flight; shared donor pools. MG63: 3 + 3; source sensitivity 3 + 2.
HMEC-1: 3 μg, 2 onboard 1g, 3 Earth cultures; its three contrasts share samples.